BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 011 · Modularity, functions & procedures

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Which loop is most suitable when the number of repetitions is known in advance?

Q2. In C#, what does && mean, and when is the whole expression true?

Q3. What is the largest unsigned 8-bit denary value?

Q4. A program reads a mark, repeats through five scores and uses if to count passes. Name the three programming constructs present.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding

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I Do · Read new knowledge

Key vocabulary

Recall the term — Modularity

Recall the term — Subroutine

Recall the term — Method

Recall the term — Function

Recall the term — Procedure

Recall the term — Return value

Recall the term — Parameter

New learning

Modularity breaks a problem into parts — A large program is easier to understand when separate jobs are placed in separate subroutines.

A procedure performs a job — A C# procedure-style method uses void because it does not return a value to the caller. — CODE:
public static void Greeting(string name)
{
Console.WriteLine("Hello " + name);
}

A function returns a value — The return type replaces void . A return statement sends the result back to the calling code. — CODE:
public static int Triple(int number)
{
return number * 3;
}

Parameters receive data — A parameter is the named input in the method definition. An argument is the actual value supplied when the method is called. — CODE:
public static int Triple(int number)
↑ parameter

Triple(7)
↑ argument

Worked example

Build an IsEven function

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

1. Choose the return type — CODE:
public static bool IsEven(int number)
{
} — The result is true/false, so the return type is bool .

2. Use the remainder test — CODE:
public static bool IsEven(int number)
{
return number % 2 == 0;
} — The Boolean expression itself already produces true or false .

3. Use it from Main — CODE:
public static void Main()
{
bool result = IsEven(12);
Console.WriteLine(result);
} — The argument 12 is received by the parameter number .

4. Use contrasting values — CODE:
IsEven(12) → True
IsEven(7) → False
IsEven(0) → True

We Do · Practise with guidance

Greeting procedure

Create a method called Greeting that receives a name and displays Hello name .

Function or procedure?

Square function

Create a method that receives an integer and returns its square.

Complete the method

```
public static ? Square(int number)
{
    ?
}

public static int Square(int number)
{
    return number * number;
}
```

Refactor repeated code

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

A program prints the same separator in several places:

Move that repeated job into one method.

Design the module

OCR asks students to distinguish the two. State the difference precisely.

Complete both statements

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

```
{
public static void Main()
{
Console.Write("First number: ");
int first = int.Parse(Console.ReadLine());

Console.Write("Second number: ");
int second = int.Parse(Console.ReadLine());

ShowLine();
int total = Add(first, second);
Console.WriteLine(total);
ShowLine();
}

public static int Add(int first, int second)
{
return first + second;
}

public static void ShowLine()
{
Console.WriteLine("--------------------");
}
}
```

Modularity: Add() has one calculation job, ShowLine() has one display job, and Main() coordinates them.

Exit Ticket · Model answer

True. A function supplies a return value to the calling code.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

```
public class Program
{
public static void Main()
{
int answer = DoubleIt(12);
Console.WriteLine(answer);
}

public static int DoubleIt(int number)
{
return number * 2;
}
}
```

Why it works: the method returns an integer, so the return type is int .

Q3. IsPass function
CODE:

```
using System;

public class Program
{
public static void Main()
{
Console.WriteLine(IsPass(65));
Console.WriteLine(IsPass(49));
}

public static bool IsPass(int mark)
{
return mark >= 50;
}
}
```

Why it works: the Boolean comparison already evaluates to true or false .

Q4. Refactor into modules
CODE:

```
using System;

public class Program
```


You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

DisplayTitle procedure
Create a procedure-style method called DisplayTitle() .
It should output OCR H446 Computer Science .
Call it from Main() twice.

Question 2

DoubleIt function
Create DoubleIt(int number) that returns twice the value supplied.
Call the function with 12 , store the returned value, then output it.

Question 3

IsPass function
Create IsPass(int mark) .
Return true when the mark is at least 50; otherwise return false .
Call it from Main() with at least two different marks.

Question 4

Refactor a program into modules
Write a program that asks for two integers and then:
1. calls Add(int first, int second) and outputs the returned total;
2. calls ShowLine() before and after the result.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Exit Ticket · Answer before checking

A function returns a value to its caller.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. A for loop . It is commonly used for count-controlled iteration.

Q2. && means AND . The whole expression is true only when both Boolean expressions are true.

Q3. 255 Eight bits give 256 patterns, representing values 0–255.

Q4. Sequence · iteration · branching/selection Instructions happen in order, a loop repeats code, and if chooses whether the pass counter changes.

You Do pages · Independent questions

Q1. DisplayTitle procedure
CODE:

```
using System;
public class Program
{
public static void Main()
{
DisplayTitle();
DisplayTitle();
}

public static void DisplayTitle()
{
Console.WriteLine("OCR H446 Computer Science");
}
}
```

Why it works: DisplayTitle() performs an action but does not return a value, so it uses void .

Q2. DoubleIt function
CODE:

```
using System;
```